Anomaly Detection in Spacecraft Telemetry using Similarity Metrics and Isolation Forest

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***Abstract* —Anomaly detection plays key role in spacecraft operations. One way of classifying anomalies is to find deviation in parameter trend by comparing it with previous data. This study aims to find such anomalies by using distance metrics like Euclidean, Cosine and using isolation forest to find out outliers in the data. Application of this method has been verified using temperature sensor data of spacecraft. To evaluate the model, outputs generated were verified with known anomalous observations and manual analysis was done for other observations. The results obtained are promising with high detection and low false alarm rates.**

Keywords—Timeseries, similarity search, telemetry, anomaly detection, isolation forest

I. Introduction

Similarity metrics gives us an insight about the extent of similarity between two or more time series data sets. Similarity search in time series data plays a major role in data mining or pattern recognition. For anomaly detection using similarity metrics, we wish to find data points that differ considerably from the desired trend. With vast amount of data at our disposal for analysis, similarity search can be used to find anomalous patterns in parameters based on desired behaviour. Similarity search methods fall mainly in to two categories, distance-based methods and data transformation methods. In distance-based methods, usage of cosine distance, Euclidean distance, Dynamic Time Warping (DTW) [1], and Pearson distance are some of the widely used techniques. In data transformation methods, usage of Discrete Fourier Transform [2], Symbolic Aggregation Approximation (SAX) [3] for the transformation of original data to an alternative representation is done for similarity search.

Master Control Facility (MCF) is responsible for monitoring and managing geosynchronous (GSO) spacecraft of Indian Space Research Organisation (ISRO). A typical GSO spacecraft has approximately 5000 telemetry parameters conveying information about various components of the spacecraft. One of the important aspects of spacecraft operations involve analysis of such telemetry channels, by comparing current data with previous data, for abnormal patterns and then plan operations to constantly meet the mission requirements which is currently done manually by a human expert. However, following such procedure for hundreds of parameters across 25-30 spacecraft becomes taxing and practically not effective which calls for a shift towards data-driven anomaly detection methods which are scalable and comprehensive.

In this paper, we define anomaly as deviation from previous data/signature. Anomaly detection is done manually by overlapping the past and current data to check deviations and all the deviations found are reported as anomalies. Automation of such process for better anomaly detection is addressed in this paper. This paper presents a novel way of detecting anomalies in huge dataset like, offline spacecraft telemetry repository, using similarity search and isolation forest [4]. It shows the application of such techniques on temperature parameters of spacecraft telemetry, further it discusses the applications and shortcoming of such methods for anomaly detection.

Future work will center on near real-time implementation, implementation of Mahalanobis distance [5] for detecting anomalies considering data distribution also and extending these techniques to multivariate approaches, exploring dimensionality reduction techniques such as Principal Component Analysis (PCA) in multivariate approach, and further refining anomaly detection algorithms to ensure robustness and scalability in real-world spacecraft operations.

II. Methods

A. Problem Statement

This study aims to find the effectiveness of similarity metrics in combination with isolation forest to find anomalies in a huge dataset like offline spacecraft telemetry. The anomaly in this case is a variation in telemetry data with

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respect to previous day data. If the variation is due to an expected operation it is still classified as anomaly only.

B. Dataset

Temperature parameters of GSO spacecraft are considered. Two years of data (01-Jan-2022 to 31-Dec-2023) sampled at every second is taken for analysis. This data also consists of a known anomalous signature and our interest is to validate it. Raw offline telemetry data is converted to engineering values using mission software tools and a program to extract day-wise data and append it to a file is used for data extraction.

C. Methodology

Initial pre-processing steps involve eliminating unwanted lines from the file and converting the dataset into a data frame. As the data is formed using a day-wise data, the duplicate time stamps were removed and null values were filled using forward-fill followed by a backward-fill technique for accurate representation. Later the data was resampled by taking a 1-minute mean to reduce the dataset. Resampling can be tuned based on data variation and its signature. The data was then separated day-wise to compare the similarity between consecutive days.

Various similarity metrics, including Euclidean distance, Cosine distance, Pearson distance, and DTW, were employed to quantify the similarity between consecutive days of telemetry data. Anomalies can be identified based on deviations in similarity metrics of consecutive days data. Thresholds for anomaly detection can be determined either manually or using the isolation forest algorithm. In this paper, the isolation forest algorithm was utilized for anomaly detection. An isolation forest classifier was then trained on selected as well as all distance values obtained, to identify anomalies. The contamination parameter of the isolation forest was tuned to adjust the sensitivity of anomaly detection.

The effectiveness of the anomaly detection approach was evaluated using historical spacecraft telemetry data. Anomalies detected by the proposed method were compared against known anomalies to assess the accuracy and reliability of the detection mechanism.

Results obtained from the anomaly detection process, including distance plots and identified anomalies, were analysed to understand the performance of each similarity metric and the overall effectiveness of the approach.

III. Results and Discussion

<u>Parameter-1 of GSO-1 (P1G1)</u>

Fig.1 to Fig. 4 represent monthly plots of Euclidean, Cosine, Pearson and DTW distances of P1G1.

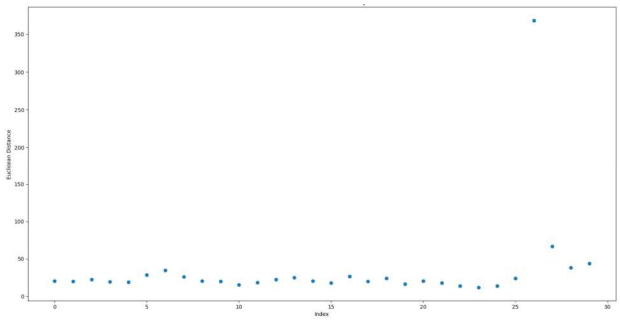

Fig. 1 Monthly plot of Euclidean distance for P1G1

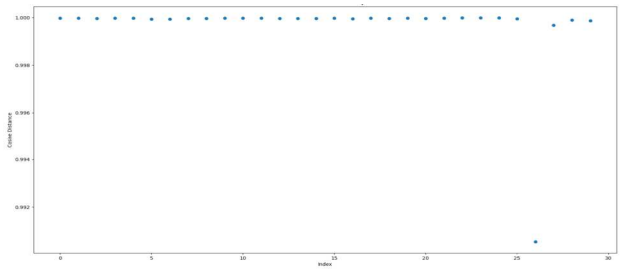

Fig. 2 Monthly plot of Cosine distance for P1G1

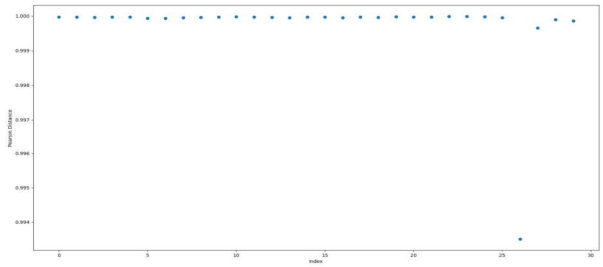

Fig. 3 Monthly plot of Pearson distance for P1G1

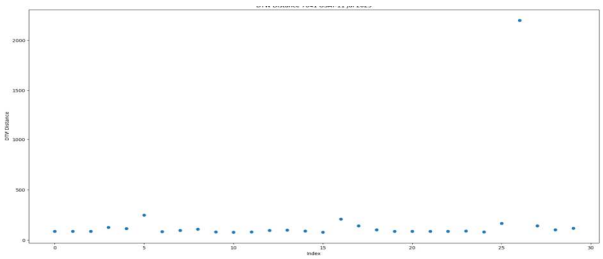

Fig. 4 Monthly plot of DTW distance for P1G1

Fig. 5 to Fig. 8 represent two-year plots of Euclidean, Cosine, Pearson and DTW distances of P1G1.

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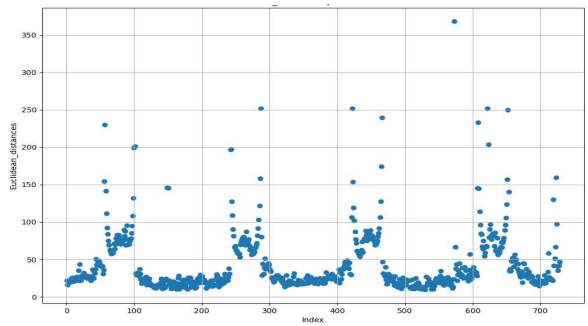

Fig. 5 Two-year plot of Euclidean distance for P1G1

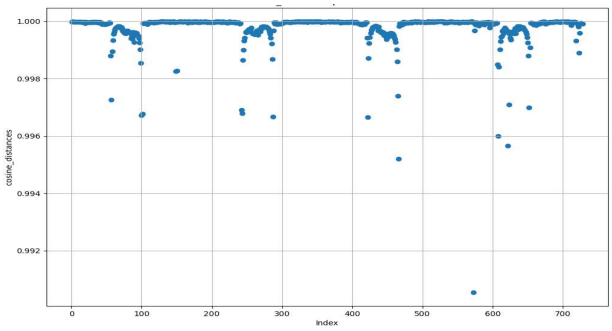

Fig. 6 Two-year plot of Cosine distance for P1G1

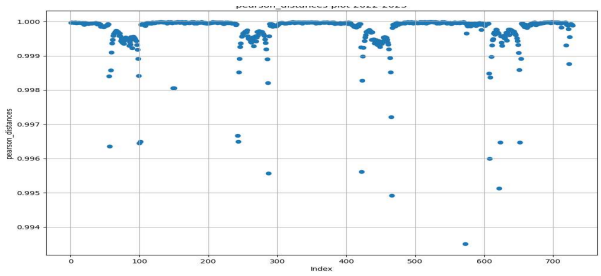

Fig. 7 Two-year plot of Pearson distance for P1G1

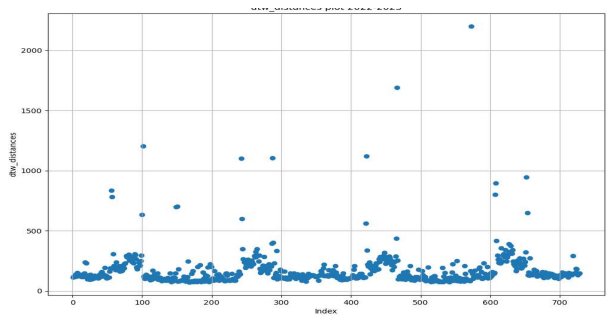

Fig. 8 Two-year plot of DTW distance for P1G1

Each data point represents the distance of day at that index with respect to day at previous index. Monthly plots shown above are taken for the month which has known anomaly and a clear outlier can be seen at the index which corresponds to anomalous date. Yearly data shows the cyclicality of temperature parameters. These data points are passed through an isolation forest algorithm with a contamination factor of 0.007 as number of reported anomalies is less in this dataset. There was only 1 known anomaly reported during this period. However, algorithm detected 6 points (including 1 known anomaly) as anomalous during this period. When the results are analyzed it was found that other 5 signatures were as per the change in seasonal trends during the period which is expected.

Fig. 9 and Fig. 10 represent plots of actual data which are detected as anomalies. The deviation in Fig. 9 is due to anomaly and in Fig. 10, it is due to seasonal behavior.

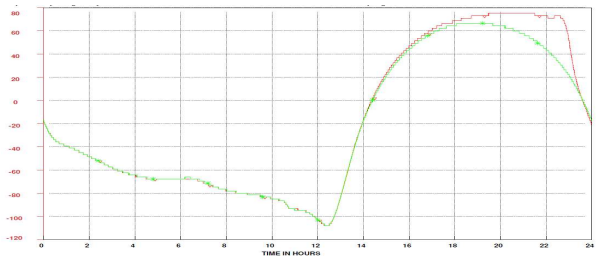

Fig. 9 Detected anomaly

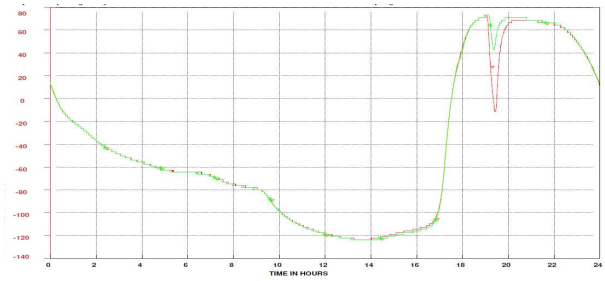

Fig. 10 Anomaly detected due to seasonal variation

Parameter – 2 of GSO – 2 (P2G2)

For this parameter experiment is repeated in 2 ways.
Case-1: Similarity between consecutive days.
Case-2: Similarity between current year data with previous year

Case-1: Fig. 11 to Fig. 14 represent two-year plots of Euclidean, Cosine, Pearson and DTW distances of P2G2.

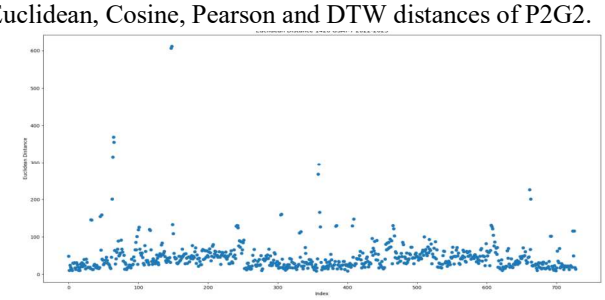

Fig. 11 Two-year plot of Euclidean distance for P2G2 (case-1)

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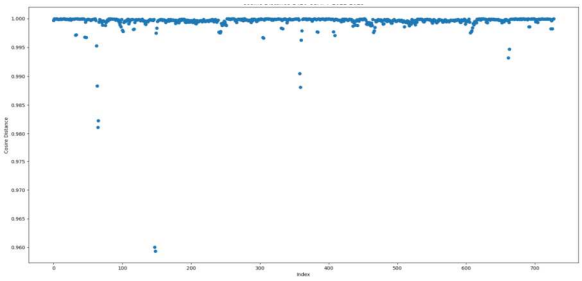

Fig. 12 Two-year plot of Cosine distance for P2G2 (case-1)

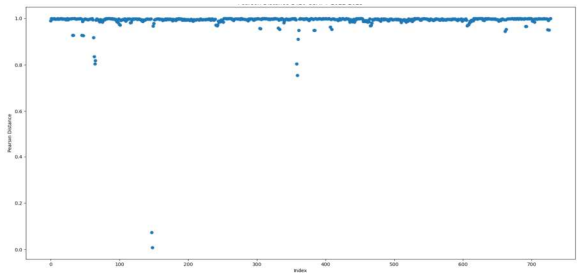

Fig. 13 Two-year plot of Pearson distance for P2G2 (case-1)

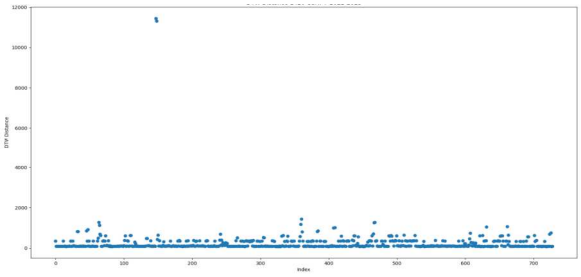

Fig. 14 Two-year plot of DTW distance for P2G2 (case-1)

When this data is given as input to an isolation forest classifier with a contamination factor of 0.007 it gave us a list of 6 possible anomalous signatures whereas during this period only 2 known anomalies are reported.

Case-2: Fig. 15 to Fig. 18 represent two-year plot of Euclidean, Cosine, Pearson and DTW distances of P2G2.

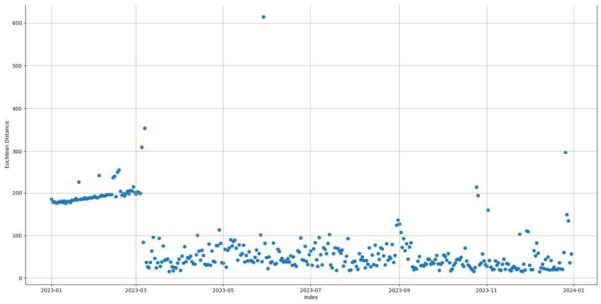

Fig. 15 Two-year plot of Euclidean distance for P2G2 (case-2)

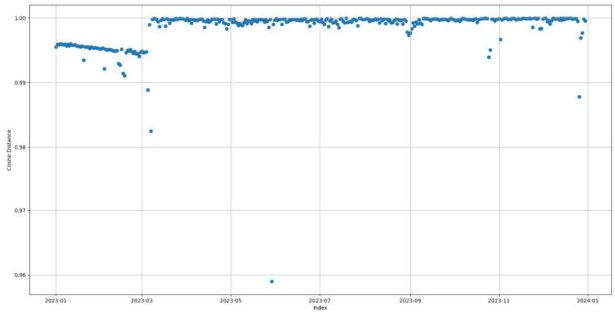

Fig. 16 Two-year plot of Cosine distance for P2G2 (case-2)

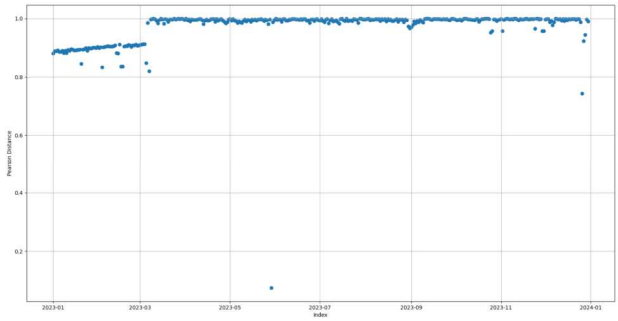

Fig. 17 Two-year plot of Pearson distance for P2G2 (case-2)

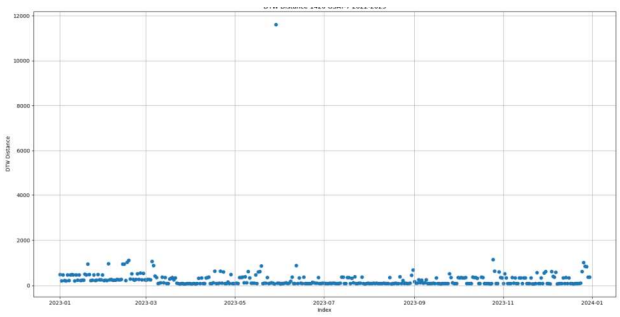

Fig. 18 Two-year plot of DTW distance for P2G2 (case-2)

There was a change in heater operation during mar 2022, so we can clearly see that when we compare data yearly it shows a jump in data distances around march segment. However, we can see that there is no significant shift in DTW since it checks for optimal alignment of data for similarity check. When the yearly comparison data is run through isolation forest algorithm with contamination factor of 0.007 it yielded us 3 possible anomalies out of which 2 are already reported as anomalies and the other 1 is because of change in heater cycling which is expected. Changing the contamination factor to 0.01 gives us 15 points as anomalous and when these are examined it was found that model becomes more sensitive to changes. So, based on experimentation and expected variations in signatures contamination factor can be fixed.

Based on the experiments conducted it can be observed that the above method can be used for anomaly detection in vast data. It can be inferred that such method is better suited for parameters whose diurnal/seasonal variations are minimal. This model fails to account for signature change

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due to known operations and signature change during seasonal changes. This can also be used to classify operations in data based on their similarity with known signatures.

IV. Conclusion

This study provides an understanding into determining an optimum solution of suitable contamination factor and distance metrics for anomaly detection in different telemetry channels using experimentation and validation. Even though the proposed method is suitable for data mining and pattern recognition in large telemetry repositories, it fails to handle pattern changes due to known operations/activities. Nevertheless, it offers robust outlier detection as the threshold is not manually fixed. This method can reduce the monitoring burden on human experts for parameters exhibiting similar diurnal trends. The key challenges of this research lie in the selection of suitable contamination factor and distance metrics for anomaly detection. By automating anomaly detection process, operational efficiency can be increased allowing spacecraft engineers to focus on critical tasks and on parameters which doesn't exhibit diurnal signatures. Anomalies detected can be further classified as operations after comparing them with labelled operational signatures and this will help in distinguishing true anomalies.

Future work can be focused on filtering anomalies into three categories: operations, known anomalies and unknown anomalies. This can also be extended to a multivariate approach along with dimensionality reduction techniques for efficient detection. Other similarity metrics like Mahalanobis distance can also be explored to take data distribution into account.

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